

1 MC Uncertainty Propagation

For a data reduction equation (DRE) of the form:

$$r = f(x_1, x_2, \dots, x_L)$$

- Generate random values for each of the inputs
- For each set of random values, evaluate random
- This generates a set of values $r = \{r_1, r_2, \dots, r_m\}$
- Then evaluate the statistics of the set r

INSERT Air diagram

After you perform uncertainty analyses on the measured T and P value, calculate an uncertainty interval for density

$$\rho = \frac{P}{R_{air}T}$$

2 MC analysis

$$r = \rho$$

$$x_1 = p$$

$$x_2 = T$$

$$x_3 = R_{air}$$

Suppose:

$$T = 298 \pm 2k(95\%)$$

$$P = \bar{P} \pm U_p k(95\%)$$

$$P = 100 \pm 1k(95\%)$$

Where:

3 code

```
m<-2e5 # of MC replicates
```

```
P <- numeric(M)
```

```
t <- numeric(M)
```

```
rho <- numeric(M)
```

```
Rair <- 287
```

Where:

$$\bar{T} = 298$$

```

for (m in 1:M) {
  p[m] <- rnorm(n=1, mean=1e5, sd = 500)
  t[m] <- rnorm(n=1, mean=298, sd = 1)
  rho[m] <- p[m]/Rair/t[m]
}
P<-0.95#conf.level
Pmin <- (1-p)/2
Pmax<-(1+p)/2
quantile(rho,probs=c(Pmin,Pmax))

```

The quantile function works on a list of numbers $\rho = \rho_1, \dots, \rho_M$ sorted from smallest to largest. For $P = 95\%$, the quantile function finds the ρ values that correspond to the upper and lower limits of 95% of the values.

```

x = {1,2,3, .. ,100}
quantile(x,probs = c(Pmin,Pmax))

```

```

Pmin=(1-p)/2 = 0.025
Pmax=(1+p)/2=0.975

```

4 Manufacturer's correlation

$$\dot{m} = c(\rho_o Z)^{1/2}$$

$$r = x_1(x_2 x_3)^{1/2}$$

Units	Quantity	Mean Value	$u = (s^2 + b^2)^{1/2}$
?	c	0.0504	0.0000005
kg/m^2	ρ_P	1.15	0.2
-	-	-	-

INSERT OBSTR Meter Theory image

5 Numerical Integration

$$\dot{m} = 2\pi \int_0^R \rho v dr$$

$$\dot{m} = 2\pi \rho \int_0^R r v dr$$

Where: $v = \left(\frac{2\Delta P}{\rho_o}\right)^{1/2}$ Bernoulli Eqn.

$$\dot{m} = 2\pi \rho \int_0^R r \left(\frac{2\Delta P}{\rho_o}\right)^{1/2} dr$$

$$\dot{m} = 2\pi \rho \left(\frac{2}{\rho_o}\right)^{1/2} \int_0^R r \Delta P^{1/2} dr$$

INSERT table and plot for delta P and r

5.0.1 algorithm

- generate a random value for ρ_o
 - `rhoD <- rnorm(n=1, mean= $\bar{\rho}_0$ , sd =  $u_{\rho_0}$ )`
- Generate one random value for each of the r values
 - ```
r1<-rnorm(n=1, mean=rbar1, sd ur1)
r2<-rnorm(n=1, mean=rbar2, sd ur2)
```
- Generate one random value for each of the  $\Delta P$ s
  - `dP1<-rnorm(n=1, mean = 54, sd =  $U_{\Delta P_1}$ )`

INSERT code photo